

Dishank Oza

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EDUCATION

Masters in Computer Science - Santa Clara University | 4.0/4.0

Sep 2024 - Mar 2026

Bachelors in Computer Engineering - University of Mumbai | 3.5/4.0

Aug 2017 - Jun 2021

Relevant Coursework: Distributed Computing, Machine Learning, Advance OS, Database Systems, Data Structures, Advanced Algorithms.

Research & Teaching Assistant (Frugal Innovation Hub|Distributed Systems): Built Airflow-orchestrated ETL pipelines exposed via FastAPI, processing **1TB** of climate data to produce **six-month forecasts** empowering farmers in Central America with data-driven planning insights.

PROFESSIONAL EXPERIENCE

Toir Inc.

Santa Clara, USA

Software Engineer III

Dec 2025 - Present

- Built a **login system** with OAuth-based authentication, **JWT-based access and refresh tokens**, session lifecycle management, and role-based access control, and owned end-to-end application containerization using Docker multi-service Compose setups, **GitHub Actions-based CI pipelines**, and AWS services to ensure consistent local, CI, and production deployments.

Teradata

San Diego, USA

Software Engineering Intern II

Jun 2025 - Sept 2025

- Upgraded Teradata's **C++ database engine** from **LLVM 16** → **20** by removing typed pointers, deprecated APIs, and redundant instruction casts, while updating **codegen/JIT interfaces** for compatibility and performance.
- Built a LLVM IR diffing and performance analysis tool for large-scale Teradata database system, debugging codegen across compiler versions, **quantifying structural shifts**, and generating visualizations, driving **~10% CPU-time reduction** on 1TB workloads.

Freespace

Mumbai, India

Senior Software Engineer

May 2023 - Aug 2024

- Led a four-engineer** Agile team to deliver key product features and streamline developer and testing **environment workflows** using Jenkins and Terraform, improving release speed, consistency, and operational efficiency — recognized as **Freespace Captain**.
- Architected a **Spring Boot microservice** leveraging **AWS DMS**, Kinesis, Lambda, and Step Functions to transition event booking from bulk pre-generation to just-in-time processing, cutting redundant computation by **60%** and enabling scalable, event-driven workflows.
- Optimized API transaction handling by replacing serialized locking with **Redis-based distributed locks** in a Spring Boot service, improving concurrency and throughput while reducing peak startup time by **70%** and easing database contention.
- Engineered a **distributed, event-driven** redesign of booking APIs using **AWS SQS FIFO**, **Lambda**, and back-pressure-aware async processing, improving resilience and throughput while cutting peak servers from **12 to 8** and infrastructure costs by **40%**.

Freespace

Mumbai, India

Software Engineer

Jun 2021 - May 2023

- Led the migration to a React-based SPA with **React Router**, **lazy loading**, route-level and custom **Webpack code splitting**, and reusable HOCs—reducing bundle size, **doubling first-load performance**, and improving repeat visit speed and user retention by **100%**.
- Developed a reusable frontend component library using **React**, **TypeScript**, and Storybook, enabling consistent branded UI patterns across applications, streamlining feature delivery, and increasing developer productivity by **20%**.
- Engineered a reusable **serverless notification system** using **AWS** (Lambda, SQS, SNS, and SES) with multi-channel delivery and CloudFormation automation, now shared across multiple products to enable scalable, fault-tolerant communication workflows.

PROJECTS

Log Aggregator and Parser (LangChain/Kafka/ELK) - ongoing

- Built a distributed tracing and observability platform for microservices with OpenTelemetry tracing, streaming traces via **Kafka**, and indexing in **Elasticsearch** for scalable querying; deployed on **K8s** with autoscaling and enhanced with **LangChain summarization**, semantic search, and natural-language debugging to accelerate incident analysis and root-cause detection.

SpaceLink (TypeScript/WebRTC/Next.js) - [Link to Project](#)

- Built a full-stack 2D virtual space platform where users customize interactive environments and engage in proximity-based video calls, using WebSockets with Protobuf for real-time avatar sync, **WebRTC** for media, and a **Node.js/Next.js**, PostgreSQL, and S3 architecture to support scalable multi-user collaboration.

TECHNICAL SKILLS

Programming: C++, JavaScript, TypeScript, Go, C, Java, Python, HTML/CSS

Databases: SQL (PostgreSQL, MySQL), NoSQL (Cassandra, MongoDB, Redis), Graph (Neo4j)

Libraries/Framework: Frontend (React.js, Next.js, Vue, Tailwind), Backend (Node.js, Spring Boot, GraphQL, Protocol Buffers, WebRTC, FastAPI)

Cloud Technologies & Tools: AWS, Firebase, Terraform, Jenkins, Git, Docker, Kubernetes/EKS, Ansible, Grafana, Prometheus, Linux, Hadoop

Development Practices: Full-stack development, CI/CD, Automation, TDD, Design Patterns, Agile, SDLC, SOLID, and Code Reviews

Certifications & Publications: [AWS Certified Developer Associate](#), Scrum Alliance-Scrum Foundations, [Gesture-Based Interaction System](#)

ACHIEVEMENTS, AWARDS, AND LEADERSHIP

- Awarded Freespace Captain for successfully leading the team, utilizing exceptional problem-solving skills, and ensuring timely delivery of project deliverables.
- Member of Team Vaayushastra, ranked 9th globally and 2nd in the Asia-Pacific region in the SAE Aero Design Competition 2020.